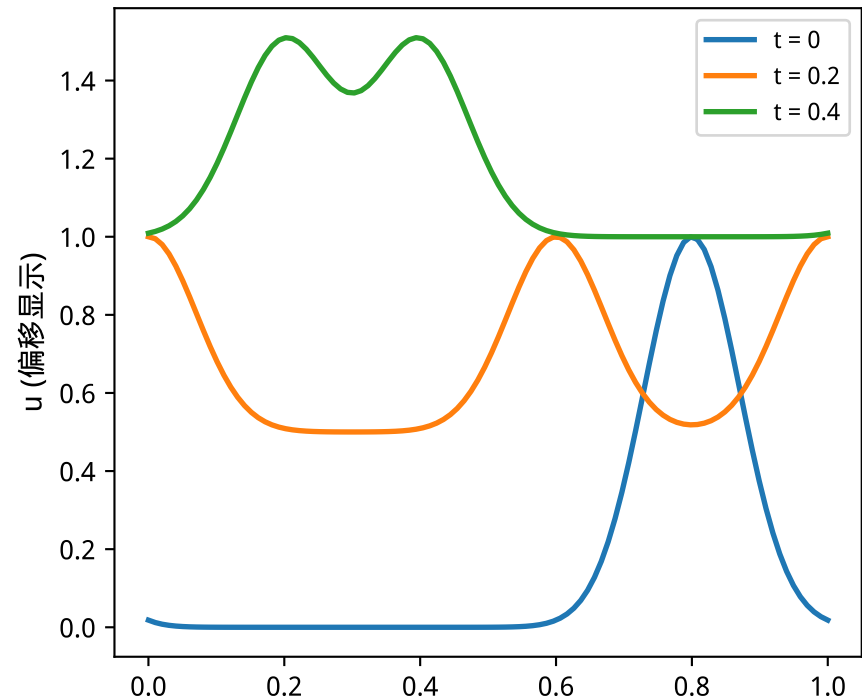


三类基本PDE的特征对比

双曲型: 波动方程

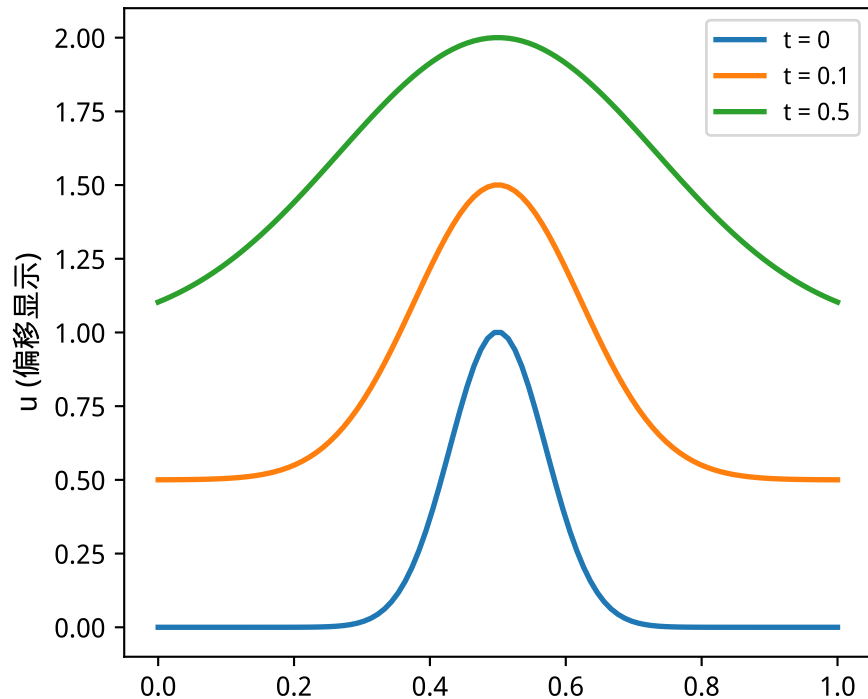
$$\partial^2 u / \partial t^2 = c^2 \partial^2 u / \partial x^2$$



特征: 波传播,
信息以有限速度传递

抛物型: 热方程

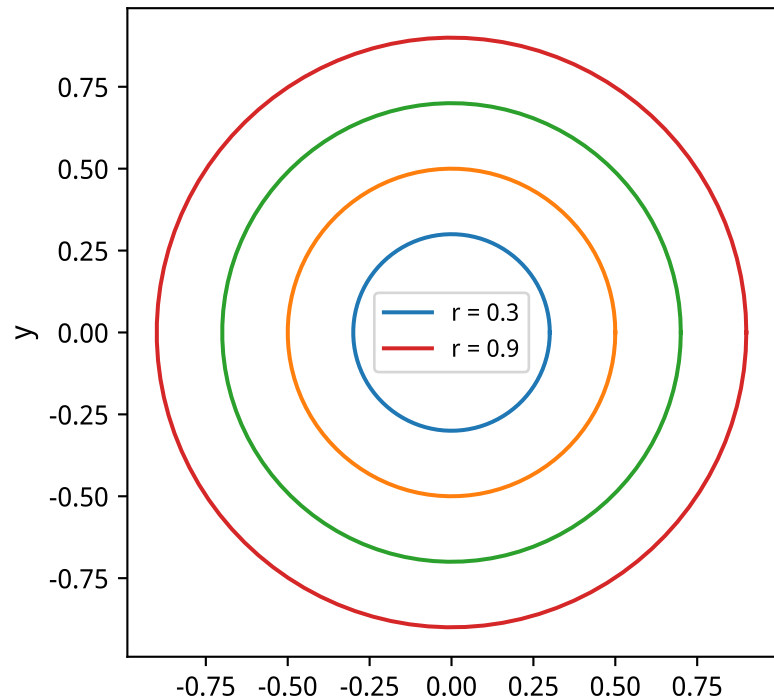
$$\partial u / \partial t = \alpha \partial^2 u / \partial x^2$$



特征: 扩散,
尖锐特征被平滑

椭圆型: Laplace方程

$$\nabla^2 u = 0$$



特征: 稳态,
边界决定内部